Knowledge Representation

Lecture 7:

First Order Predicate Calculus

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First–Order Predicate Calculus FOPC

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- a quantifier $\forall$
- parentheses (,).

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$$\begin{aligned} \alpha \vee \beta &\stackrel{def}{=} \neg\alpha \to \beta \\ \alpha \vee \beta &\stackrel{def}{=} \neg(\neg\alpha \wedge \neg\beta) \\ \alpha \equiv \beta &\stackrel{def}{=} (\alpha \to \beta) \wedge (\beta \to \alpha) \end{aligned}$$

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$$\alpha \equiv \beta \quad \stackrel{def}{=} \quad (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$$

- ***Existential quantifier***: $\exists x.\, \alpha(x) \stackrel{def}{=} \neg\forall x.\, \neg\alpha(x)$.

Terms

A ***term*** over Al is an expression of the form:

- each variable $x \in Var$ is a term
- if $\tau_1, \ldots, \tau_n$ are terms, $n \geq 0$, and $f^{(n)} \in \mathcal{F}$ is a function symbol of the arity n, then $f^{(n)}(\tau_1, \ldots, \tau_n)$ is a term
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Example 5.4 x, $x + y$, $father(x)$, $Child(John, Mary)$.

Formulas of FOPC

An ***atomic formula*** (atom) is an expression of the form:

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A ***formula*** is an expression of the form:

- each atom is a formula
- if α and β are formulas, then so are $\neg\alpha$, $\alpha \wedge \beta$, $\alpha \vee \beta$, $\alpha \rightarrow \beta$, $\alpha \equiv \beta$, $\forall x.\alpha$
- no other expressions are formulas.

Formulas of FOPC (cont.)

Example 7.1

$$Student(Brother(x))$$
$$\forall x, y.\ Likes(x, y) \wedge HasBirthday(y) \rightarrow GivesPresent(x, y).$$

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$\forall x.\exists y.\ Likes(x, y)$	closed formula
$\forall x.\ IsFriend(x, John) \rightarrow Likes(x, y)$	open formula.

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A ***theory*** of FOPC is a set of closed formulas.

Representation issues

Example 7.3 Consider the following sentences:

- *All students are adults.*
- *Some children do not learn English*
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$\exists x. Child(x) \wedge \neg LearnEnglish(x).$

$\forall x.\ StudentCS(x) \rightarrow (\exists y.\ Examines(x, y) \wedge ReasercherCS(y)).$

Semantics of FOPC

Let $\mathcal{L}$ be a language of FOPC. A ***frame*** for $\mathcal{L}$ is a structure $\boxed{\boldsymbol{\Phi = (D, m)}}$ where

- D is a nonempty domain
- m is a mapping which assigns:
  - to each function symbol $f \in \mathcal{F}$ of arity n, a mapping $m(f) : D^n \to D$
  - to each predicate symbol $R \in \mathcal{R}$ of arity n, a relation $m(R) \subseteq D^n$.

Valuation of terms

Let $\Phi = (D, m)$ be a frame for $\mathcal{L}$.

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  The set of all assignments over Φ will be denoted by $As(\Phi)$.

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E.g. for $D = \mathbb{N}$, $+$ interpreted as usual addition, and a given by $a(x) = 5$, $a(y) = 7$, we have:

$$val_a^\Phi(x + y) = val_a^\Phi(x) + val_a^\Phi(y) = 5 + 7 = 12.$$

Valuation of formulas

Let $\Phi = (D, m)$ be a frame for $\mathcal{L}$, $a \in As(\phi)$, and let $\alpha \in \mathcal{L}$.

Denote: $a_x = \{a' \in As(\Phi) : a(y) = a'(y)$ for any $y \neq x\}$.

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Satisfiability

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Accordingly, α is ***unsatisfiable*** iff it has no model.

Axiomatization of FOPC

Logical axioms:

- $\top$
- $\alpha \to (\beta \to \alpha)$
- $(\alpha \to (\beta \to \gamma)) \to (\alpha \to \beta) \to (\alpha \to \gamma)$
- $(\alpha \to \beta) \to ((\alpha \to \neg\beta) \to \neg\beta)$
- $\forall x.\, (\alpha \to \beta) \to (\alpha \to \forall x.\, \beta)$, provided that there is no free occurrences of x in α
- $\forall x.\, \alpha(x) \to \alpha(\tau)$, where τ is free for x in $\alpha(x)$.

E.g. y is free for x in $P(x)$, but if not free for x in $\exists y.\, P(x, y)$.

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Reasoning rules

- **(MP)** $\dfrac{\alpha, \alpha \to \beta}{\beta}$
- **(GEN)** $\dfrac{\alpha}{\forall x.\, \alpha}$ where x is not free in α.

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- **Theorem 7.5 (Compactness)** T is consistent iff each finite subtheory of T is consistent.
- **Theorem 7.6** $\{\alpha_1, \ldots, \alpha_n\} \models \beta$ iff $\models \alpha_1 \wedge \ldots \wedge \alpha_n \to \beta$
- **Theorem 7.7** $T \models \alpha$ iff $T \cup \{\neg\alpha\}$ is unsatisfiable.

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Theorem 7.8 The validity problem for FOPC is not decidable, but **partly decidable**.

In view of the above theorem, there exists an algorithm such that

- for every tautology it returns the answer **YES**,
- for some formula, which is not a tautology, it runs into infinite loop.

Some examples

Example 7.4: Tweety

Consider the following sentences:

- *Tweety is a bird.*
- *All birds can fly.*
- *Everyone who can fly has wings.*

Show that from these sentences it follows:

“Tweety has wings.”

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$$\beta = HasWings(Tweety).$$

Example 7.4 (cont.)

Notation: For $\Phi = (D, m)$ and any $R^{(n)} \in \mathcal{R}$,

$$|R^{(n)}|_\Phi = \{(d_1, \ldots, d_n) : d_i \in D, i = 1, \ldots, n, (d_1, \ldots, d_n) \in m(R^{(n)})\}.$$

$|R^{(n)}|_\Phi$ – an ***extension*** of $R^{(n)}$ in Φ.

Example 7.4 (cont.)

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First we show that $T = \{\alpha_1, \alpha_2, \alpha_3\}$ has a model.

Take $\Phi = (D, m)$ such that

- $D = \{t\}$
- $|Bird|_\Phi = |Flies|_\Phi = |HasWings|_\Phi = \{t\}$.

Example 7.4 (cont.)

$\Phi = (D, m)$ such that $D = \{t\}$ and
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- $m(Tweety) = t$. $t \in |Bird|_\Phi$, so $\Phi \models \alpha_1$.

Example 7.4 (cont.)

$\Phi = (D, m)$ such that $D = \{t\}$ and
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- $As(\Phi) = \{a\}$, where $a : x \mapsto t$.

Example 7.4 (cont.)

$\Phi = (D, m)$ such that $D = \{t\}$ and
$|Bird|_\Phi = |Flies|_\Phi = |HasWings|_\Phi = \{t\}$.

- $m(Tweety) = t$. $t \in |Bird|_\Phi$, so $\Phi \models \alpha_1$.
- $As(\Phi) = \{a\}$, where $a : x \mapsto t$.
- $val_a^\Phi(\forall x.\, Bird(x) \to Flies(x)) = 1$, because $t \in |Bird|_\Phi$ and $t \in |Flies|_\Phi$.
  Since a is the only assignment, $\Phi \models \alpha_2$.

Example 7.4 (cont.)

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- $val_a^\Phi(\forall x.\, Flies(x) \to HasWings(x)) = 1$, because $t \in |Flies|_\Phi$ and $t \in |HasWings|_\Phi$.
  Similarly, $\Phi \models \alpha_3$.

Example 7.4 (cont.)

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  Since a is the only assignment, $\Phi \models \alpha_2$.
- $val_a^\Phi(\forall x. Flies(x) \to HasWings(x)) = 1$, because $t \in |Flies|_\Phi$ and $t \in |HasWings|_\Phi$.
  Similarly, $\Phi \models \alpha_3$.

Example 7.4 (cont.)

Let $\Phi = (D, m)$ be an arbitrary model of $T = \{\alpha_1, \alpha_2, \alpha_3\}$. Then

- the function m assigns some object $d \in D$ to the individual constant $Tweety$.

Example 7.4 (cont.)

Let $\Phi = (D, m)$ be an arbitrary model of $T = \{\alpha_1, \alpha_2, \alpha_3\}$. Then

- the function m assigns some object $d \in D$ to the individual constant $Tweety$.
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Example 7.4 (cont.)

Let $\Phi = (D, m)$ be an arbitrary model of $T = \{\alpha_1, \alpha_2, \alpha_3\}$. Then

- the function m assigns some object $d \in D$ to the individual constant $Tweety$.
- $\Phi \models \alpha_1$, so $d \in |Bird|_\Phi$.
- $\Phi \models \alpha_2$, so for every object $d' \in D$ such that $d' \in |Bird|_\Phi$, it holds $d' \in |Flies|_\Phi$. In particular, since $d \in |Bird|_\Phi$, we get $d \in |Flies|_\Phi$.

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- Analogously, since $\Phi \models \alpha_3$, $d \in |HasWings|_\Phi$.
- Therefore, $\Phi \models \beta$.

Example 7.4 (cont.)

Let $\Phi = (D, m)$ be an arbitrary model of $T = \{\alpha_1, \alpha_2, \alpha_3\}$. Then

- the function m assigns some object $d \in D$ to the individual constant $Tweety$.
- $\Phi \models \alpha_1$, so $d \in |Bird|_\Phi$.
- $\Phi \models \alpha_2$, so for every object $d' \in D$ such that $d' \in |Bird|_\Phi$, it holds $d' \in |Flies|_\Phi$. In particular, since $d \in |Bird|_\Phi$, we get $d \in |Flies|_\Phi$.
- Analogously, since $\Phi \models \alpha_3$, $d \in |HasWings|_\Phi$.
- Therefore, $\Phi \models \beta$.
- Since Φ was an arbitrary model of T, we get that $T \models \beta$.

Example 7.5: Boring novels

Consider the following sentences:

- *Some people like all novels.*
- *Nobody likes boring novels.*

Show that from these sentences it follows

"*There are no boring novels*".

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Representation in FOPC:

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$$\alpha_1 = \exists x.\ Man(x) \wedge (\forall y.\ Novel(y) \to Likes(x, y))$$

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$$\beta = \forall x.\ Novel(x) \rightarrow \neg Boring(x).$$

Example 7.5 (cont.)

We show that $T = \{\alpha_1, \alpha_2\}$ has a model. Take $\Phi = (D, m)$ such that

- $D = \{A, B\}$
- $|Man|_\Phi = \{A\}$
- $|Likes|_\Phi = \{(A, B)\}$.
- $|Boring|_\Phi = \emptyset$
- $|Novel|_\Phi = \{B\}$

Example 7.5 (cont.)

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$$\alpha_1 = \exists x.\ Man(x) \land (\forall y.\ Novel(y) \rightarrow Likes(x, y))$$

$\Phi \models \alpha_1$, because there is an object $d \in D$ such that

- $d \in |Man|_\Phi$ — indeed, $A \in |Man|_\Phi$.
- for every object $d' \in D$ such that $d' \in |Novel|_\phi$, it holds $(d, d') \in |Likes|_\Phi$.
  Indeed, $B \in |Novel|_\Phi$ and $(A, B) \in |Likes|_\Phi$. B is the only object from D which belongs to $|Novel|_\Phi$.

Example 7.5 (cont.)

- $D = \{A, B\}$
- $|Man|_\Phi = \{A\}$
- $|Likes|_\Phi = \{(A, B)\}$.
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Example 7.5 (cont.)

- $D = \{A, B\}$
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$$\alpha_2 = \forall x.\, \forall y.\, Man(x) \wedge Novel(y) \wedge Boring(y) \rightarrow \neg Likes(x, y)).$$

$\Phi \models \alpha_2$, because

- $|Boring|_\Phi = \emptyset$, so there is no object $d' \in D$, such that $d \in |Novel|_\Phi$ and $d \in |Boring|_\Phi$ and $d' \in |Man|_\Phi$. Hence, the left side of the implication is false, so the whole implication is true.

Example 7.5 (cont.)

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- $|Man|_\Phi = \{A\}$
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Hence, $\Phi \models T$.

Example 7.5 (cont.)

$$\alpha_1 = \exists x.\ Man(x) \wedge (\forall y.\ Novel(y) \rightarrow Likes(x, y))$$
$$\alpha_2 = \forall x.\ \forall y.\ Man(x) \wedge Novel(y) \wedge Boring(y) \rightarrow \neg Likes(x, y))$$
$$\beta = \forall x.\ Novel(x) \rightarrow \neg Boring(x).$$

Suppose $T \not\models \beta$. From the theorem

$$T \models \beta \quad \text{iff} \quad T \cup \{\neg\beta\} \text{ unsatisfiable}$$

it holds $T \cup \{\neg\beta\}$ has a model, say $\Phi = (D, m)$.

Example 7.5 (cont.)

$$\alpha_1 = \exists x.\ Man(x) \wedge (\forall y.\ Novel(y) \to Likes(x,y))$$
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$\neg\beta = \exists x.\ Novel(x) \wedge Boring(x)$.

$\Phi \models \neg\beta$ iff there is $d_0 \in D$ such that

(1) $d_0 \in |Novel|_\Phi$

(2) $d_0 \in |Boring|_\Phi$.

Example 7.5 (cont.)

$$\alpha_1 = \exists x.\ Man(x) \wedge (\forall y.\ Novel(y) \rightarrow Likes(x, y))$$
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$$\beta = \forall x.\ Novel(x) \rightarrow \neg Boring(x).$$

$\Phi = (D, m)$ and there is $d_0 \in D$ such that

(1) $d_0 \in |Novel|_\Phi$ **(2)** $d_0 \in |Boring|_\Phi$

Example 7.5 (cont.)

$$\alpha_1 = \exists x.\ Man(x) \wedge (\forall y.\ Novel(y) \rightarrow Likes(x, y))$$
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$\Phi = (D, m)$ and there is $d_0 \in D$ such that

(1) $d_0 \in |Novel|_\Phi$ **(2)** $d_0 \in |Boring|_\Phi$

$T \models \alpha_1$ iff there is $d_1 \in D$ such that

(3) $d_1 \in |Man|_\Phi$

(4) for every $d \in D$, $d \in |Novel|_\Phi \Longrightarrow (d_1, d) \in |Likes|_\Phi$.

(1) and **(4)** implies

(5) $(d_1, d_0) \in |Likes|_\Phi$

Example 7.5 (cont.)

$$\alpha_1 = \exists x.\ Man(x) \wedge (\forall y.\ Novel(y) \rightarrow Likes(x, y))$$
$$\alpha_2 = \forall x.\ \forall y.\ Man(x) \wedge Novel(y) \wedge Boring(y) \rightarrow \neg Likes(x, y))$$
$$\beta = \forall x.\ Novel(x) \rightarrow \neg Boring(x).$$

$\Phi = (D,)$ and there are $d_0, d_1 \in D$ such that

(1) $d_0 \in |Novel|_\Phi$

(2) $d_0 \in |Boring|_\Phi$

(3) $d_1 \in |Man|_\Phi$

(4) for every $d \in D$,
$d \in |Novel|_\Phi \Longrightarrow (d_1, d) \in |Likes|_\Phi$

(5) $(d_1, d_0) \in |Likes|_\Phi$

Example 7.5 (cont.)

$$\alpha_1 = \exists x.\, Man(x) \wedge (\forall y.\, Novel(y) \to Likes(x,y))$$
$$\alpha_2 = \forall x.\, \forall y.\, Man(x) \wedge Novel(y) \wedge Boring(y) \to \neg Likes(x,y))$$
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(3) $d_1 \in |Man|_\Phi$

(4) for every $d \in D$, $d \in |Novel|_\Phi \Longrightarrow (d_1, d) \in |Likes|_\Phi$

(5) $(d_1, d_0) \in |Likes|_\Phi$

$T \models \alpha_2$ iff for all $d, d' \in D$,

$$d \in |Man|_\Phi \ \ \&\ \ d' \in |Novel|_\Phi \ \ \&\ \ d' \in |Boring|_\Phi \Longrightarrow (d, d') \notin |Likes|_\Phi$$

so by **(1)**, **(2)**, **(3)**, $(d_1, d_0) \notin |Likes|_\Phi$ – a contradiction with **(5)**.

Example 7.6: Barber's Paradox

Consider the following sentences:

- *All barbers shave all men who do not shave themselves.*
- *No barber shaves a man who shaves himself.*

Show that these sentences imply:

"There is no barber.

Example 7.6: Barber's Paradox

Consider the following sentences:

- *All barbers shave all men who do not shave themselves.*
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Representation in FOPC:

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Representation in FOPC:

$$\alpha_1 = \forall x, y.\ Barber(x) \wedge \neg Shaves(y, y) \rightarrow Shaves(x, y)$$

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Representation in FOPC:

$$\alpha_1 = \forall x, y.\ Barber(x) \wedge \neg Shaves(y, y) \rightarrow Shaves(x, y)$$

$$\alpha_2 = \forall x, y.\ Barber(x) \wedge Shaves(y, y) \rightarrow \neg Shaves(x, y)$$

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$$\alpha_2 = \forall x, y.\ Barber(x) \wedge Shaves(y, y) \rightarrow \neg Shaves(x, y)$$
$$\beta = \forall x.\ \neg Barber(x).$$

Example 7.6 (cont.)

$$\alpha_1 = \forall x, y.\ Barber(x) \wedge \neg Shaves(y, y) \rightarrow Shaves(x, y)$$
$$\alpha_2 = \forall x, y.\ Barber(x) \wedge Shaves(y, y) \rightarrow \neg Shaves(x, y).$$

Note that $\Phi = (D, m)$ such that

- $D = \{b\}$
- $|Barber|_\Phi = \emptyset$
- $|Shaves|_\Phi = \{(b, b)\}$.

Example 7.6 (cont.)

$$\alpha_1 = \forall x, y.\ Barber(x) \land \neg Shaves(y, y) \to Shaves(x, y)$$
$$\alpha_2 = \forall x, y.\ Barber(x) \land Shaves(y, y) \to \neg Shaves(x, y)$$
$$\beta = \forall x.\ \neg Barber(x).$$

Suppose that $\{\alpha_1, \alpha_2\} \not\models \beta$, i.e. $\{\alpha_1, \alpha_2, \neg\beta\}$ has a model, say $\Phi = (D, m)$.

$\Phi \models \neg\beta$ iff there is $d_0 \in D$ such that

(1) $d_0 \in |Barber|_\Phi$.

$\Phi \models \alpha_1$ iff for all $d, d' \in D$

(2) $d \in |Barber|_\Phi$ & $(d', d') \notin |Shaves|_\Phi \Longrightarrow (d, d') \in |Shaves|_\Phi$.

Example 7.6 (cont.)

$$\alpha_1 = \forall x, y.\ \mathit{Barber}(x) \wedge \neg \mathit{Shaves}(y, y) \rightarrow \mathit{Shaves}(x, y)$$
$$\alpha_2 = \forall x, y.\ \mathit{Barber}(x) \wedge \mathit{Shaves}(y, y) \rightarrow \neg \mathit{Shaves}(x, y)$$
$$\beta = \forall x.\ \neg \mathit{Barber}(x).$$

$\Phi = (D, m)$ and there is $d_0 \in D$ such that

(1) $d_0 \in |\mathit{Barber}|_\Phi$.

(2) for all $d, d' \in D$,
$d \in |\mathit{Barber}|_\Phi$ & $(d', d') \notin |\mathit{Shaves}|_\Phi \Longrightarrow (d, d') \in |\mathit{Shaves}|_\Phi$.

(1) and **(2)** imply

(3) for any $d' \in D$, $(d', d') \notin |\mathit{Shaves}|_\Phi \Longrightarrow (d_0, d') \in |\mathit{Shaves}|_\Phi$.

From **(1)** and **(3)**, $(d_0, d_0) \notin |\mathit{Shaves}|_\Phi \Longrightarrow (d_0, d_0) \in |\mathit{Shaves}|_\Phi$, so

(4) $(d_0, d_0) \in |\mathit{Shaves}|_\Phi$.

Example 7.6 (cont.)

$$\alpha_1 = \forall x, y.\ \mathit{Barber}(x) \wedge \neg \mathit{Shaves}(y, y) \rightarrow \mathit{Shaves}(x, y)$$
$$\alpha_2 = \forall x, y.\ \mathit{Barber}(x) \wedge \mathit{Shaves}(y, y) \rightarrow \neg \mathit{Shaves}(x, y)$$
$$\beta = \forall x.\ \neg \mathit{Barber}(x).$$

$\Phi = (D, m)$ and there is $d_0 \in D$ such that

(1) $d_0 \in |\mathit{Barber}|_\Phi$

(4) $(d_0, d_0) \in |\mathit{Shaves}|_\Phi$

$\Phi \models \alpha_2$ iff for all $d, d'' \in D$,

(5) $d \in |\mathit{Barber}|_\Phi$ & $(d'', d'') \in |\mathit{Shaves}|_\Phi \Longrightarrow (d, d'') \notin |\mathit{Shaves}|_\Phi$

(1), **(4)**, and **(5)** imply $(d_0, d_0) \notin |\mathit{Shaves}|_\Phi$ – a contradiction with **(4)**.

Example 7.7: Drug pushers

Consider the following sentences:

- *The custom officials search everybody who enters the country and is not a VIP.*
- *Some drug pushers, who enter the county, are searched only by drug pushers.*
- *No drug pusher is a VIP.*

Show that these sentences imply:

“*Some custom officials are drug pushers.*”

Example 7.7: Representation

Denote:

- $E(x)$ – x enters the country.
- $V(x)$ – x is a VIP
- $C(x)$ – x is a custom official.
- $P(x)$ – x is a drug pusher.
- $S(x, y)$ – x searches y.

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- $C(x)$ – x is a custom official.
- $P(x)$ – x is a drug pusher.
- $S(x, y)$ – x searches y.

The custom officials search everybody who enters the country and is not a VIP.

$$\forall x.\, E(x) \land \neg V(x) \rightarrow (\exists y.\, C(y) \land S(y, x))$$

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- $P(x)$ – x is a drug pusher.
- $S(x,y)$ – x searches y.

The custom officials search everybody who enters the country and is not a VIP.

$$\forall x.\ E(x) \wedge \neg V(x) \rightarrow (\exists y.\ C(y) \wedge S(y,x))$$

Some drug pushers, who enter the county, are searched only by drug pushers.

$$\exists x.\ P(x) \wedge E(x) \wedge (\forall y.\ S(y,x) \rightarrow P(y))$$

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- $P(x)$ – x is a drug pusher.
- $S(x, y)$ – x searches y.

The custom officials search everybody who enters the country and is not a VIP.

$$\forall x.\ E(x) \land \neg V(x) \to (\exists y.\ C(y) \land S(y, x))$$

Some drug pushers, who enter the county, are searched only by drug pushers.

$$\exists x.\ P(x) \land E(x) \land (\forall y.\ S(y, x) \to P(y))$$

No drug pusher is a VIP. $\forall x.\ P(x) \to \neg V(x)$.

Example 7.7: Representation

Denote:

- $E(x)$ – x enters the country.
- $V(x)$ – x is a VIP
- $C(x)$ – x is a custom official.
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The custom officials search everybody who enters the country and is not a VIP.

$$\forall x.\ E(x) \wedge \neg V(x) \rightarrow (\exists y.\ C(y) \wedge S(y, x))$$

Some drug pushers, who enter the county, are searched only by drug pushers.

$$\exists x.\ P(x) \wedge E(x) \wedge (\forall y.\ S(y, x) \rightarrow P(y))$$

No drug pusher is a VIP. $\forall x.\ P(x) \rightarrow \neg V(x)$.

Some custom officials are drug pushers. $\exists x.\ C(x) \wedge P(x)$.

Example 7.7 (cont.)

$$\alpha_1 = \forall x.\ E(x) \wedge \neg V(x) \rightarrow (\exists y.\ C(y) \wedge S(y,x))$$
$$\alpha_2 = \exists x.\ P(x) \wedge E(x) \wedge (\forall y.\ S(y,x) \rightarrow P(y))$$
$$\alpha_3 = \forall x.\ P(x) \rightarrow \neg V(x)$$
$$\beta = \exists x.\ C(x) \wedge P(x)$$

Proceeding as before it is easy to show that $T = \{\alpha_1, \alpha_2, \alpha_3\}$ has a model.

Let $\Phi = (D, m)$ be such a model.

$\Phi \models \alpha_2$, so there is $d_0 \in D$ such that

(1) $d_0 \in |E|_\Phi$

(2) $d_0 \in |P|_\Phi$

(3) for every $d' \in D$, $(d', d_0) \in |S|_\Phi \Longrightarrow d' \in |D|_\Phi$.

$\Phi \models \alpha_3$, so

(4) $d_0 \notin |V|_\Phi$.

Example 7.7 (cont.)

$$\begin{aligned}\alpha_1 &= \forall x.\ E(x) \wedge \neg V(x) \to (\exists y.\ C(y) \wedge S(y,x))\\ \alpha_2 &= \exists x.\ P(x) \wedge E(x) \wedge (\forall y.\ S(y,x) \to P(y))\\ \alpha_3 &= \forall x.\ P(x) \to \neg V(x)\\ \beta &= \exists x.\ C(x) \wedge P(x).\end{aligned}$$

$\Phi = (D, m)$ and there is $d_0 \in D$ such that

(1) $d_0 \in |E|_\Phi$
(2) $d_0 \in |P|_\Phi$
(3) for every $d' \in D$, $(d', d_0) \in |S|_\Phi \Longrightarrow d' \in |D|_\Phi$
(4) $d_0 \notin |V|_\Phi$.

$\Phi \models \alpha_1$, so by 1) and (4) there is $d_1 \in D$ such that

(5) $d_1 \in |C|_\Phi$
(6) $(d_1, d_0) \in |S|_\Phi$.

(3) and **6)** imply $d_1 \in |P|$, which together with **(5)** gives $T \models \beta$.

Thank you for your attention!

Any questions are welcome.